Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026

Mid-Semester Test
(EC-2 Regular)

Course No	:	IS ZC444
Course Title	:	Artificial Intelligence
Nature of Exam	:	Closed Book
Weightage	:	30%
Duration	:	2 hours
Date	:	21/09/2025

No. of Pages = 3
No. of Questions = 4

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1. **[3 marks]**

PEAS specification (Intelligent Agents). For each scenario below, give a concise PEAS description: list 2–3 clear items under each of **P**erformance, **E**nvironment, **A**ctuators, **S**ensors. Keep items specific to the task.

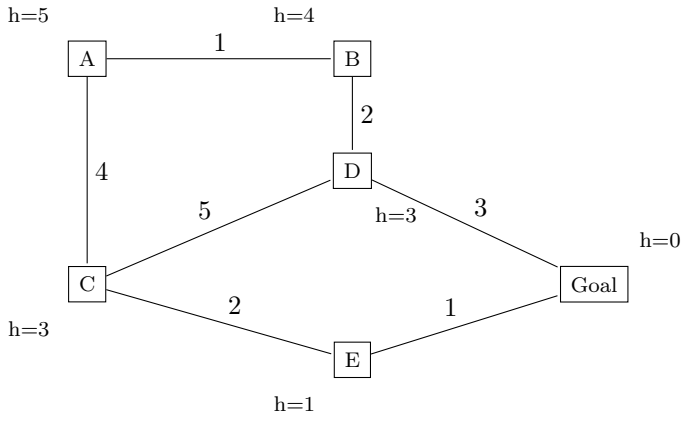
(a) *Autonomous taxi (ride-hailing) operating in a modern city.* (2 marks)

(b) Name any *two* key environment properties for the autonomous taxi task (choose from: fully/partially observable, deterministic/stochastic, episodic/sequential, static/dynamic, discrete/continuous, single-agent/multiagent) with a one-phrase justification each. (1 mark)

Q.2. **[9 marks]**

A* Search Algorithm (Informed Heuristic Search Strategies).

Consider the following weighted graph, where the numbers on the edges represent the cost of the path between the nodes, and the numbers near the nodes represent the heuristic values (h) of the nodes (to the goal).



(a) Explain the concept of the A* search algorithm. What is the evaluation function used in A*, and how does it help in finding the optimal path? (2 marks)

(b) Calculate the values of **g(n)** and **f(n)** for all the nodes that are added to the priority queue. (3 marks)

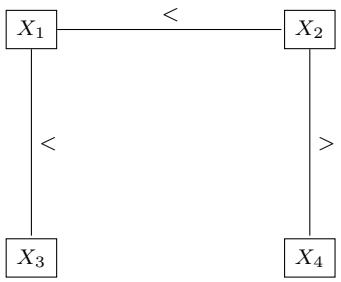
(c) Identify the sequence of nodes that A* pops out of the queue before reaching the goal. Explain why A* chooses the node to expand based on the lowest **f(n)** value. (3 marks)

(d) What is the significance of the heuristic function in the A* search algorithm? How does a consistent and admissible heuristic affect the algorithm's performance? (1 mark)

Q.3. **[9 marks]**

Constraint Satisfaction Problem (CSP) with Binary Constraints.

Consider the following CSP where the variables X_1, X_2, X_3, X_4 are assigned integer values from the domain $\{0, 1, 2, 3\}$. The binary constraints between the variables are as follows:



The constraints between the variables are: $X_1 < X_2$, $X_2 > X_4$, and $X_1 < X_3$.

Each variable can take integer values in the range $\{0, 1, 2, 3\}$.

(a) What is the meaning of the sentence given below? The goal of a constraint satisfaction problem is to find a *consistent* and *complete* assignment. (2 marks)

(b) What is node consistency and arc consistency? (2 marks)

(c) Suppose we use the AC-3 (Arc-Consistency) algorithm, to enforce arc consistency on X_1, X_2, X_3, X_4. What will be the domain of each variable when the AC-3 algorithm terminates? (4 marks)

(d) How does the AC-3 algorithm help in solving CSPs efficiently? (1 mark)

Q.4. **[9 marks]**

Knowledge Representation and Entailment in Propositional Logic.

Consider the propositional statements given below:

- p: The student attended the lecture.
- q: The student submitted the assignment.
- r: The student received a grade of 'A'.

The following three propositions represent the student's behavior and is present in the knowledge base KB:

1. If the student attended the lecture, they[1] submitted the assignment.

[1] *They* is a gender-neutral singular pronoun.

2. If the student submitted the assignment, they received a grade of 'A'.
3. If the student did not receive a grade of 'A', they did not attend the lecture.

Now, answer the following:

(a) Express the knowledge base above in terms of propositional logic. Clearly state the set of propositions that represent the knowledge about the student's behavior. (3 marks)

(b) When can we say that a knowledge base KB entails a sentence α (i.e. $KB \models \alpha$)? (2 marks)

(c) Find the models that satisfy the knowledge base given above. Each model in your answer should clearly show the truth value assignments to symbols — e.g. $(p = true, q = false, r = false)$. (2 marks)

(d) Suppose we add the following fact to the KB about a student: (2 marks)

- The student submitted the assignment

Is it logically entailed that the student attended the lecture? Justify your answer by using the definition of entailment.